The AI Security Layers
Modern Teams Cannot Skip

AI Security Layers	Guards	Attack	Stack	Core job
Data Governance and Lineage	Training and pipeline data	Poisoned training data (LLM04)		Classify and trace data flow
Sensitive Data and PII Filtering	What goes in and comes out	Leaked sensitive info (LLM02)	NEXUS	Redact on the fly
Model Supply Chain Checks	External weights and artifacts	Compromised dependencies (LLM03)		Scan for tampered models
Model Access Control	Endpoints and API keys	Unauthorized model use		Grant least privilege
Prompt Injection Defense	User and external input	Jailbreaks and override (LLM01)		Screen hostile input
Behavioral Guardrails	How the model responds	Leaked system prompts (LLM07)		Hold topic and policy lines
Output Validation	Generated responses	Mishandled output (LLM09)		Verify schema and content
Secrets and Key Handling	Credentials and tokens	Reused or leaked keys		Vault and rotate keys
RAG and Vector Protection	Embeddings and retrieval	Weak vector boundaries (LLM08)		Keep tenants isolated
Agent and Tool Sandboxing	Actions and tool calls	Too much agency (LLM06)		Limit what agents can do
Workload and Agent Identity	Machine and agent credentials	Hijacked workload identity		Issue short-lived identity
Encryption Everywhere	Data at rest, transit, in use	Interception and theft	TLS	Encrypt and isolate in enclaves
Runtime Monitoring	Live traffic and behavior	Drift and live abuse	arize	Flag anomalies early
Cost Abuse Controls	Inference spend and quotas	Unbounded usage (LLM10)		Cap spikes and throttle abuse
Adversarial Red Teaming	Pre-launch week spots	Hidden vulnerabilities	Garak, PyRIT	Attack it before others do
Threat Modeling and Response	The full attack surface	No plan when breached		Map risks and ready response
Compliance and Audit Trail	Controls and evidence	Failed audits and fines	aws	Map controls to frameworks